AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 7 · STUDENT LESSON

Categories and Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Define the label. Then inspect the evidence.

Replace a vague word, trace the calendar policy, compare craft cards and transfer the method to a new record.

Keep one habit: Cite the criterion → keep failure separate from unknown → record the policy version.

MAIN LESSON · SECTION 1

What evidence earns a label?

> Apply a shared policy, including its boundaries and missing-information rule.

Start with an ambiguous word

Mina and Theo both label a card “good”, but they may mean different things. Before comparing labels, agree on the observations that make a card belong to a category.

Retrieve the missing-word limit

A transcript leaves out an unclear word. Why should a later system not simply treat that missing word as a known value?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can apply a **category** definition using explicit **evidence** and boundary rules.
- I can distinguish a known failure from information that is missing or unreadable.
- I can use **disagreement** to improve a **label**ling guide without mistaking agreement for truth.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

Labels are human choices that may **overlap** or be unclear.
</details>



LEARNING TARGET Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can apply a shared **label**ling policy consistently, including missing information.

MAIN LESSON · SECTION 2

Replace “good” with observable checks

> The classroom calendar policy checks a record, not the worth of its author or event.

Read the disagreement

Two students are sorting fictional event cards for a classroom calendar. Their first instruction is Put the good cards together. Mina calls a colourful card good. Theo calls a plain card good because its room is easy to read.

They disagree even though both followed a reasonable interpretation of the word good.

Define the task using printed fields

A **category** is useful for a task when people can tell what **evidence** makes an example belong to it. An **operational definition** states the **observation**s or checks to perform.

For this paper exercise, we will check two printed fields: an event day and a room code. The task is to check the record against a classroom policy, not to judge the quality of the event or the person who wrote it.

Connect the guide to training labels

Labels depend on the definition in use. A model fitted to **label**s can learn the distinctions those **label**s encode, including mistakes and inconsistencies.

Before asking a model to copy a **label**, ask whether people have a clear, useful guide for assigning it.

Explain the ambiguity

- What does Mina mean by good? What does Theo mean? Which printed fields will our new policy check, and what claim about an event would those checks still leave unproved?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A playground object can be both red and round, a toy and sports equipment, or safe in one situation and unsafe in another.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Check known failures before unknown fields

> Follow the policy’s priority and include both range endpoints.

Read the complete calendar policy

Our fictional calendar accepts a whole-number day from 1 through 30, including both endpoints, and exactly one room code, A or B. The month is already fixed for this exercise. A blank or unreadable field counts as unknown.

The categories are revise, review and ready. Use the checks in the order shown.

Trace the three branches

Three ordered checks for a calendar card

CHECK IN THIS ORDER	EVIDENCE	RECORDED LABEL
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Distinguish revise from review

Known failure and unknown are different. Day 31 is known and outside the accepted range, so the card needs revision. An unreadable day might be valid or invalid; without another known failure, it needs review.

Never silently turn a blank into zero just to make a comparison possible.

Explain one deciding branch

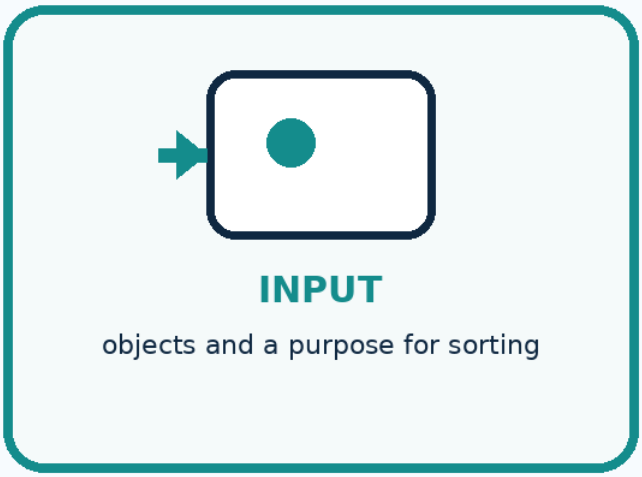
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

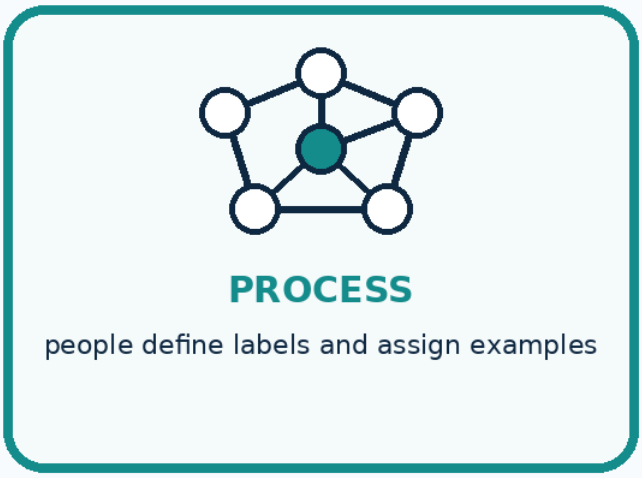
VISUAL **MECHANISM**

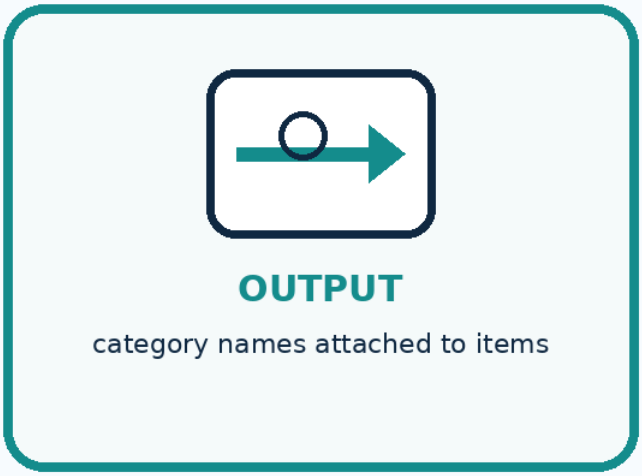
Input → Process → Output

Follow the information instead of imagining magic inside the machine.



</details>





INPUT objects and a purpose for sorting

PROCESS people define **label**s and assign examples

OUTPUT category names attached to items

CORE IDEA **Label**s are human choices that may **overlap** or be unclear.

UNDER THE HOOD A **label** is an **operational definition** chosen for a task. Categories may be mutually exclusive, multi-**label**, hierarchical, subjective, or context-dependent.

Ambiguity in **label**s becomes **ambiguity** in **training** and evaluation.

Draw a fresh example with all three stages:

A closer explanation

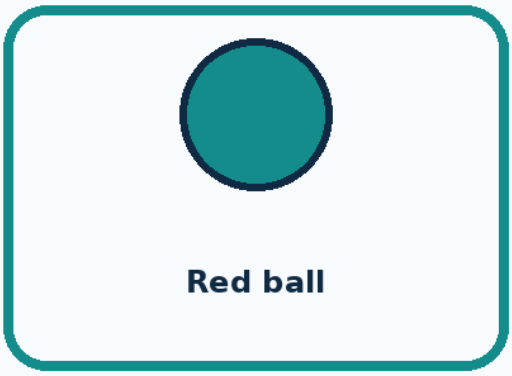
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare category clues

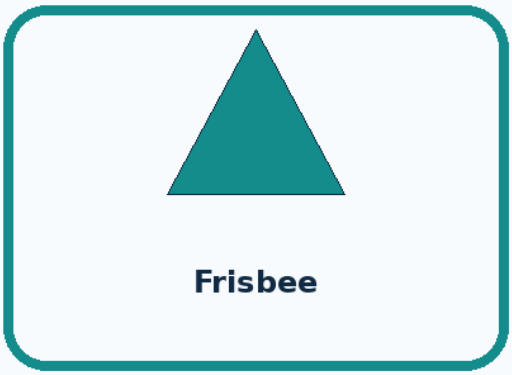
> Use the stated definition and ask which evidence is missing or could support overlapping labels.

Pictures paired with their evidence



The case to investigate

CASE 1 — Red ball Clue: red and round. Meaning: multiple descriptive **label**s.



The case to investigate

CASE 2 — Frisbee Clue: toy or sports equipment. Meaning: purpose changes **label**.



The case to investigate

CASE 3 — Wet slide Clue: equipment and possible hazard. Meaning: context matters.



The case to investigate

CASE 4 — Mystery object Clue: insufficient features. Meaning: need **more information**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

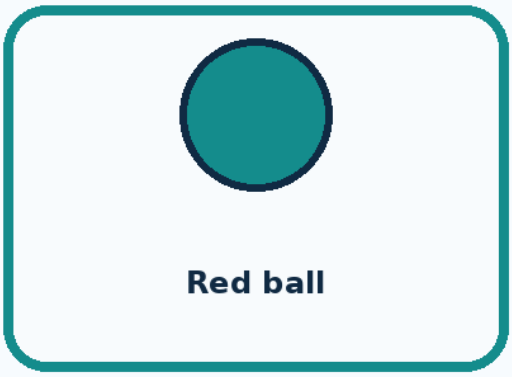
BOUNDARY **Label**s do not exist naturally in the data; people define them for a purpose.

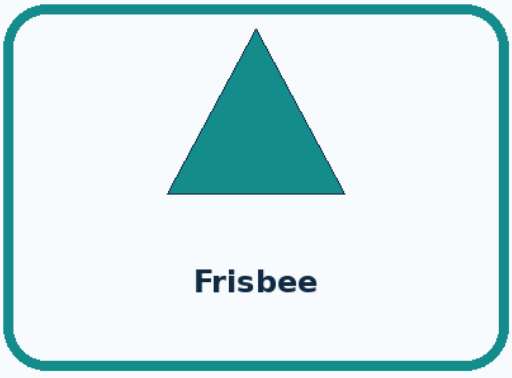
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Red ball Clue: red and round. Meaning: multiple descriptive **label**s.

CASE 2 — Frisbee Clue: toy or sports equipment. Meaning: purpose changes **label**.

CASE 3 — Wet slide Clue: equipment and possible hazard. Meaning: context matters.

CASE 4 — Mystery object Clue: insufficient features. Meaning: need **more information**.

Change one thing in the pictured case

Change: Change B's quantity from 5 to 4; leave colour blank.

Prediction: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

One policy. Three different next steps.

> Use the craft-club quantity and colour facts without importing the calendar’s ranges.

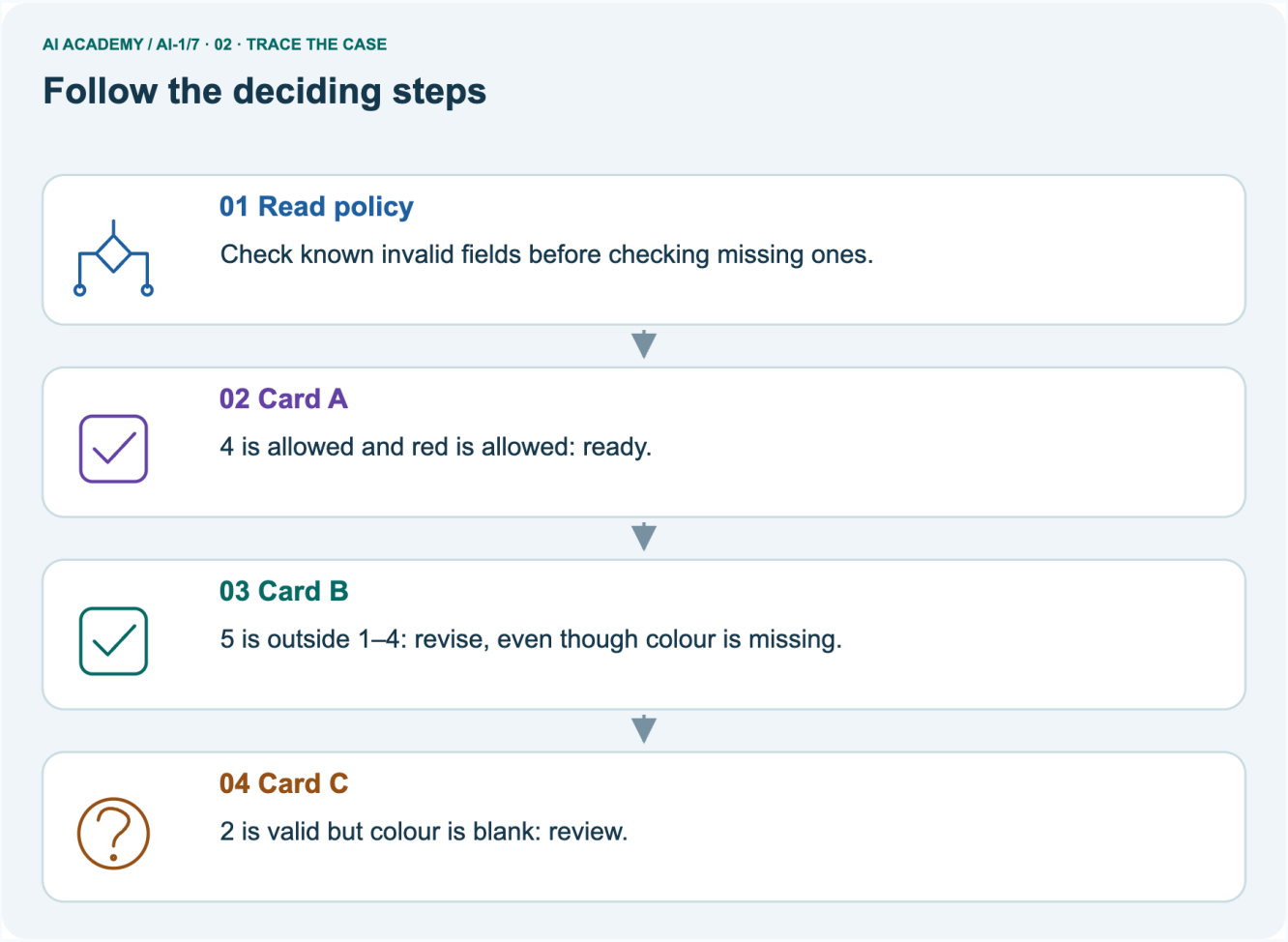
The craft-club facts and priority

A fictional craft club **label**s cardboard-order cards ready, review or revise.

Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

Trace each card’s deciding check



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the resulting labels

AI ACADEMY / AI-1/7 · 03 · INSPECT THE LESSON MODEL

Three ordered checks for a calendar card

Check in this order	Evidence	Recorded label
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the known-failure priority

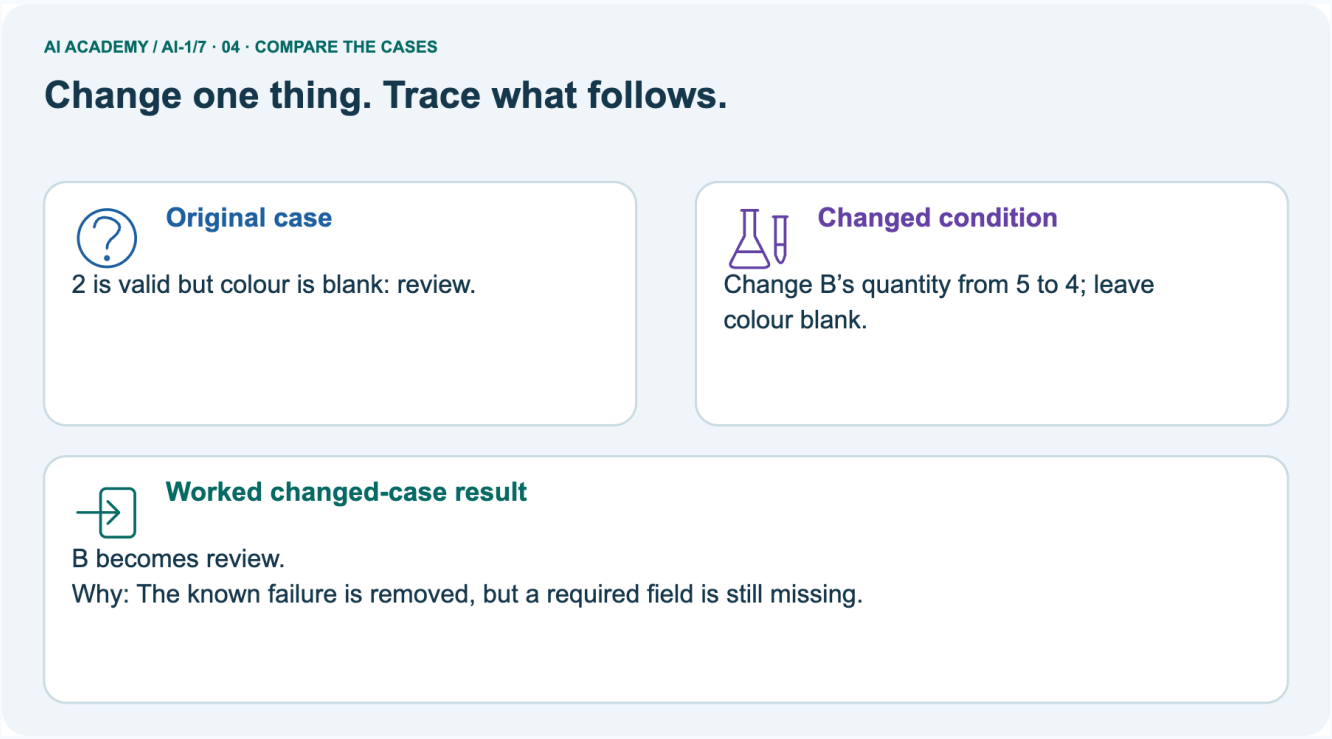
Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure.

Predict the repaired quantity

- Before checking, change B's quantity from 5 to 4 and leave its colour blank. Apply the same checks in the same order. Which known failure disappeared, and which uncertainty remains?

▼ Check the changed label



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep agreement separate from policy quality

Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

Explain the original and changed B

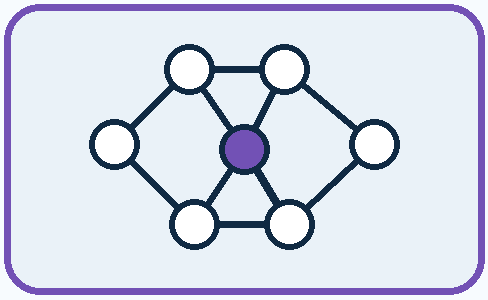
- Read the **label**led picture: Explain why B originally was revise rather than review, then **label** its changed version.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A playground object can be both red and round, a toy and sports equipment, or safe in one situation and unsafe in another.

STEP 1 — OBSERVE objects and a purpose for sorting

STEP 2 — TRACE people define **label**s and assign examples

STEP 3 — CONCLUDE **category** names attached to items

MODEL REASONING Strong **label**s include a purpose and a rule for membership.
Overlapping objects may need multiple **label**s or a priority rule.

A revised definition should reduce **disagreement** without pretending every case is simple.

DO NOT OVERCLAIM **Label**s do not exist naturally in the data; people define them for a purpose.

What new **evidence** could change this answer?

A closer explanation

Card fields	Relevant evidence	Label and reason
Day 12 room A	Both values satisfy the criteria	Ready because both checks pass
Day 31 room A	31 is outside 1 through 30	Revise because a known value fails
Day 12 room unreadable	No known failure but the room is unknown	Review until the room is checked

Card fields	Relevant evidence	Label and reason
Day 31 room unreadable	The day already fails although the room is unknown	Revise because the first check has priority

For the last card, first read only what is known: day 31. Compare it with the allowed range and record a failure. The guide therefore selects revise before the missing-room check.

Checking the room later could reveal another problem, but it cannot make day 31 satisfy this policy.

A clear explanation cites the particular criterion. Saying It looks wrong is not enough. Write Day 31 exceeds the accepted maximum of 30 instead.

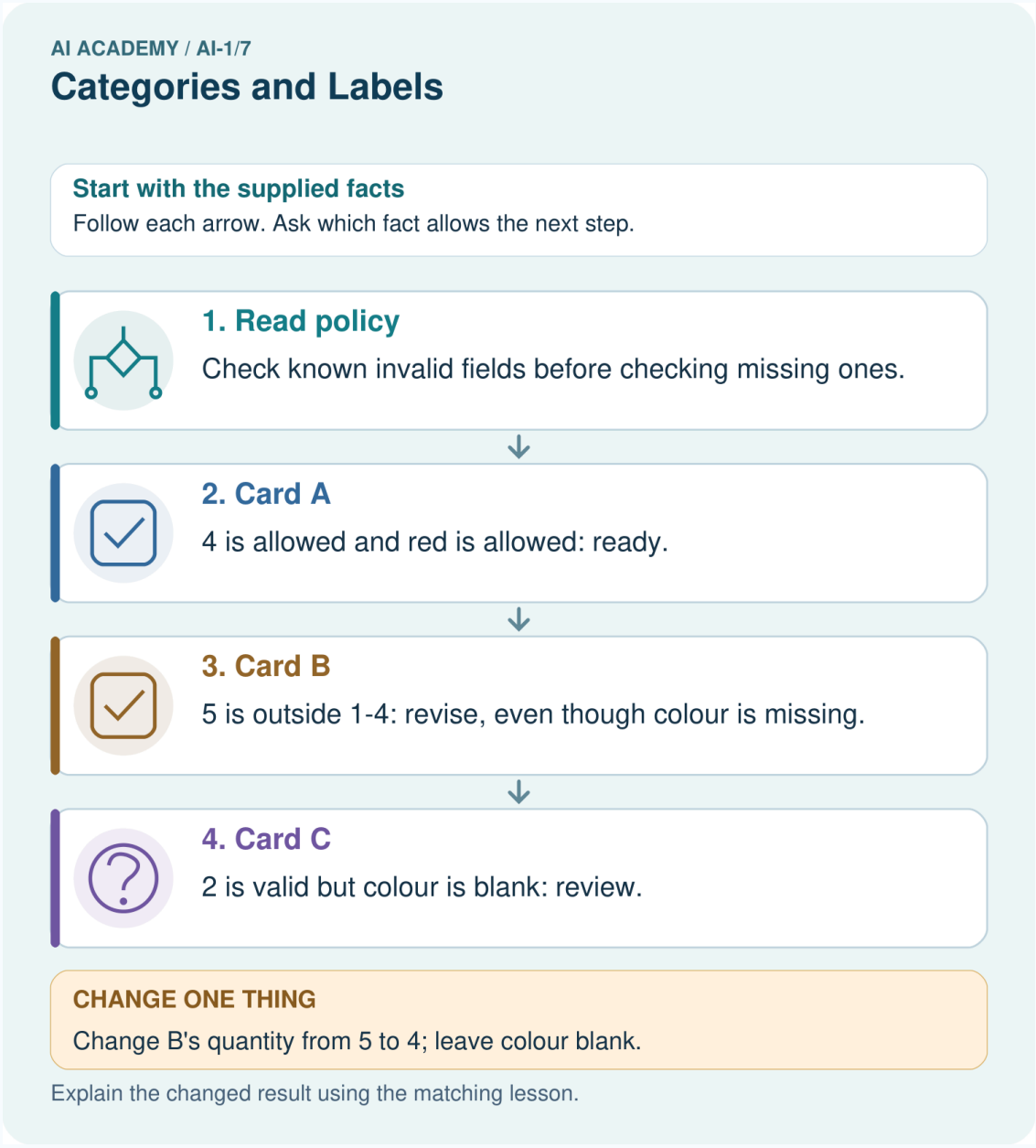
That explanation lets another person reproduce the check and challenge it if the recorded value or policy is different.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Categories and Labels

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read policy	Check known invalid fields before checking missing ones.
2. Card A	4 is allowed and red is allowed: ready.
3. Card B	5 is outside 1–4: revise, even though colour is missing.
4. Card C	2 is valid but colour is blank: review.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Change B's quantity from 5 to 4; leave colour blank. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Agreement can include shared mistakes

> Check labels against the guide and keep the policy version with the record.

Four agreements out of six: what is known?

If two **annotator**s agree on four of six cards, we know that four assigned **label**s matched. We do not yet know whether those **label**s followed the guide. Both **annotator**s might have made the same mistake.

Check their reasoning against the criteria and, where necessary, the source record.

A field check does not confirm the real event

Even perfect agreement under our calendar policy would not prove that the room is really booked. It would only show agreement about whether the printed fields pass the specified checks.

The policy itself may need changes for a real calendar. Record the policy version so examples **label**led under different instructions are not mixed without review.

Repair and explain

- CLAIM A Every object has exactly one correct **label**.
- CLAIM B **Disagreement** proves one person was careless.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **Two people label every card ready. Explain why their agreement alone does not establish correct labels or confirm the events**

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Red ball	red and round	multiple descriptive labels
Frisbee	toy or sports equipment	purpose changes label
Wet slide	equipment and possible hazard	context matters
Mystery object	insufficient features	need more information

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: defend the boundary with the guide

> Keep original independent labels before discussing a disagreement.

Resolve day 30 using the stated endpoint

Mina **label**s day 30, room B as ready; Theo **label**s it revise because he thought the range stopped before 30. The **disagreement** points to a boundary interpretation. The guide says including both endpoints, so 30 passes.

The appropriate repair is to clarify and practise the inclusive boundary, not to change the **label** merely because one student is more confident.

> **Label day 0 with room A and day 18 with a blank room. Cite the exact rule for each and explain why the two cards need different next steps**

Compare after both learners have tried

Ask **annotator**s to **label** examples independently before comparing. If they copy one another, agreement no longer tells us whether the guide can be applied independently.

Keep the original **label**s and reasons when discussing differences, then record any corrected **label**s separately.

Use W2 to replace a vague instruction

Use workbook W2 for the connected example “Categories and operational **label**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “blank means failing”

- Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the changed craft card

Explain why B originally was revise rather than review, then **label** its changed version.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Label**s are human choices that may **overlap** or be unclear.

Use a fresh example to explain **category**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Disagreement** proves one person was careless.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Replace a vague instruction

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate: write a guide before comparing labels

> Use supplied fictional fields so overlap, unknown information and a changed definition can be checked.

Cards and complete shared policy

Use four fictional playground-inventory cards. A: shape=sphere, opening=no. B: shape=box, opening=yes. C: shape=sphere, opening=yes. D: shape=unreadable, opening=no. Our paper policy checks both fields first: if either is unreadable, record REVIEW. Otherwise assign SPHERE when shape=sphere and OPEN when opening=yes. Both labels may apply; if neither applies, record OTHER.

Predict, compare and revise the guide

1. Write the two label definitions before labelling. Predict each result independently, then compare with a partner. Cite the field behind each label and keep C's overlap visible. For D, request the shape information instead of treating unreadable as box. Propose a different useful grouping rule and record its version before trying it on the same cards.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep original labels and reasons

Case	Prediction	Observed result	What it means
Normal			
Edge			

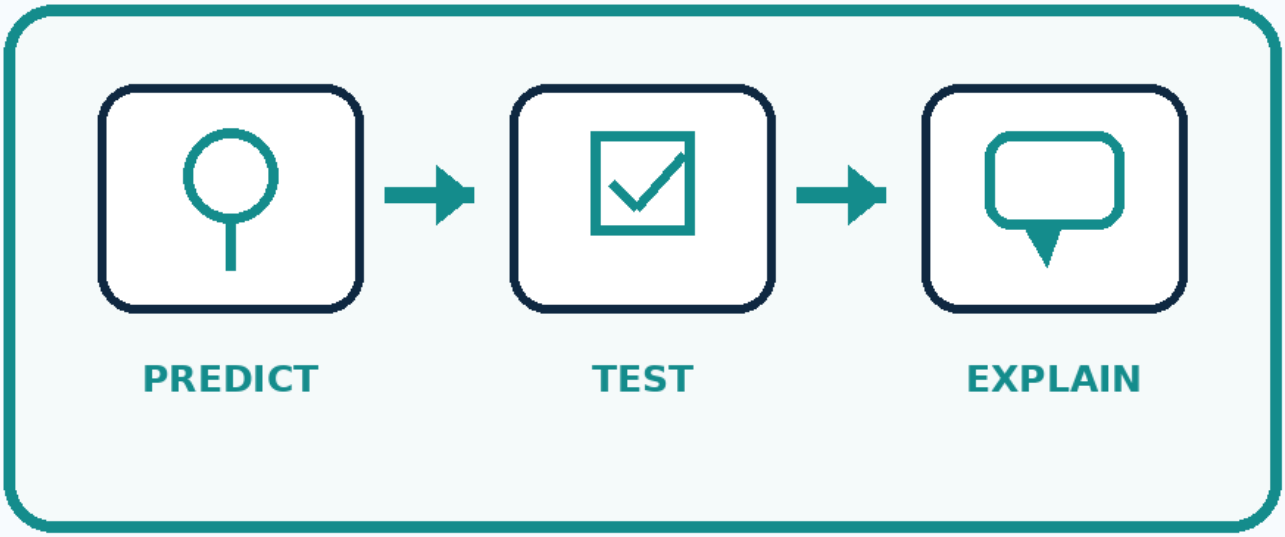
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Playground Label Council

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Design **label**s for playground objects; challenge ambiguous cards.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: transfer the method to equipment forms

> Use the new duration range, item codes and priority exactly as supplied.

The new policy and forms

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Transfer the method to a new record

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For this toy sorter, label an object Seed-bearing when seeds are observed, Other when inspection establishes that it has no seeds, and Review when the needed contents cannot be observed. Sort an opened apple with visible seeds, a pencil and a sealed mystery box. Which item needs more **evidence**, and why?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Give your sorter a versioned label guide

> Write the criteria and how to handle invalid, unknown or overlapping cases.

This week’s project decision

THIS WEEK'S CHECKPOINT Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

Plan the evidence and next action

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep the task-dependent boundary

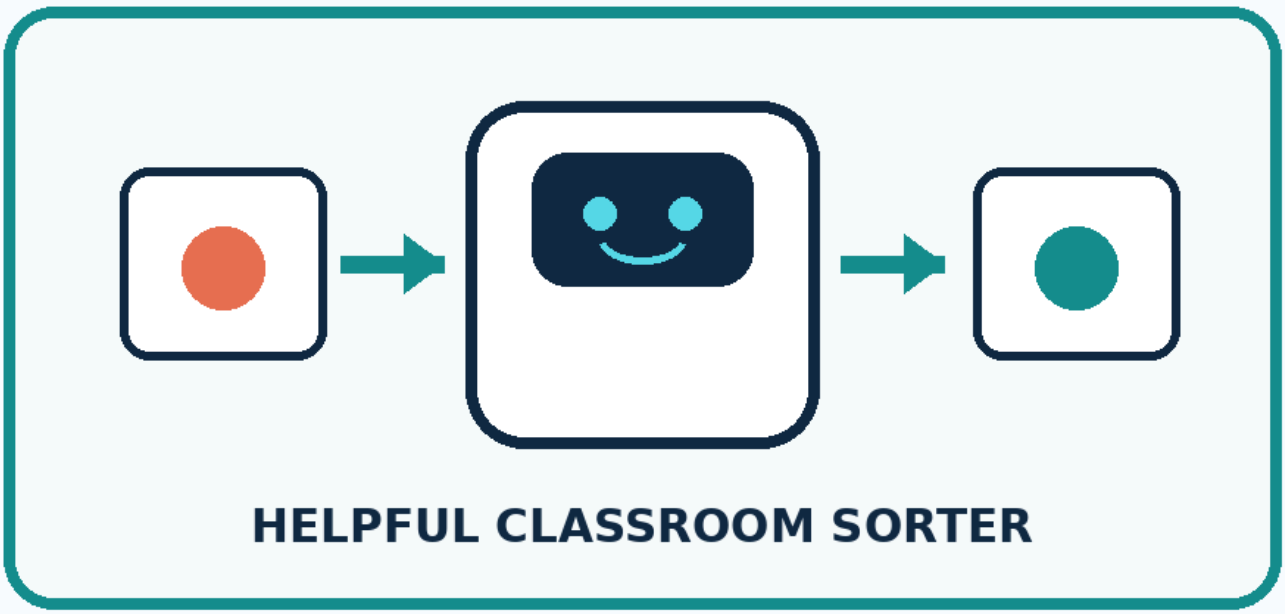
RESPONSIBLE DESIGN **Label**s do not exist naturally in the data; people define them for a purpose.

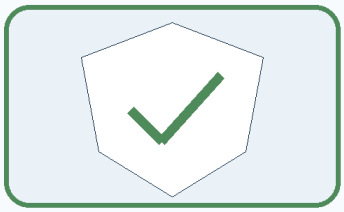
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify a label with its criterion

> Close your notes and explain the difference between known failure and unknown information.

Your exit explanation

- Close your notes. Explain this goal in your own words: Apply a **category** definition using explicit **evidence** and boundary rules

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
category	group created for a purpose	
label	name assigned to an example	
overlap	when more than one label can fit	
ambiguity	when evidence allows multiple reasonable meanings	

Terms for the connected explanation

Word	Meaning
Operational definition	A definition precise enough to tell someone what to observe or do when assigning a category
Annotator	A person who assigns labels to examples

Word	Meaning
Disagreement	A case where annotators assign different labels to the same example

Repair a claim using a word

- CLAIM TO REPAIR Every object has exactly one correct **label**.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

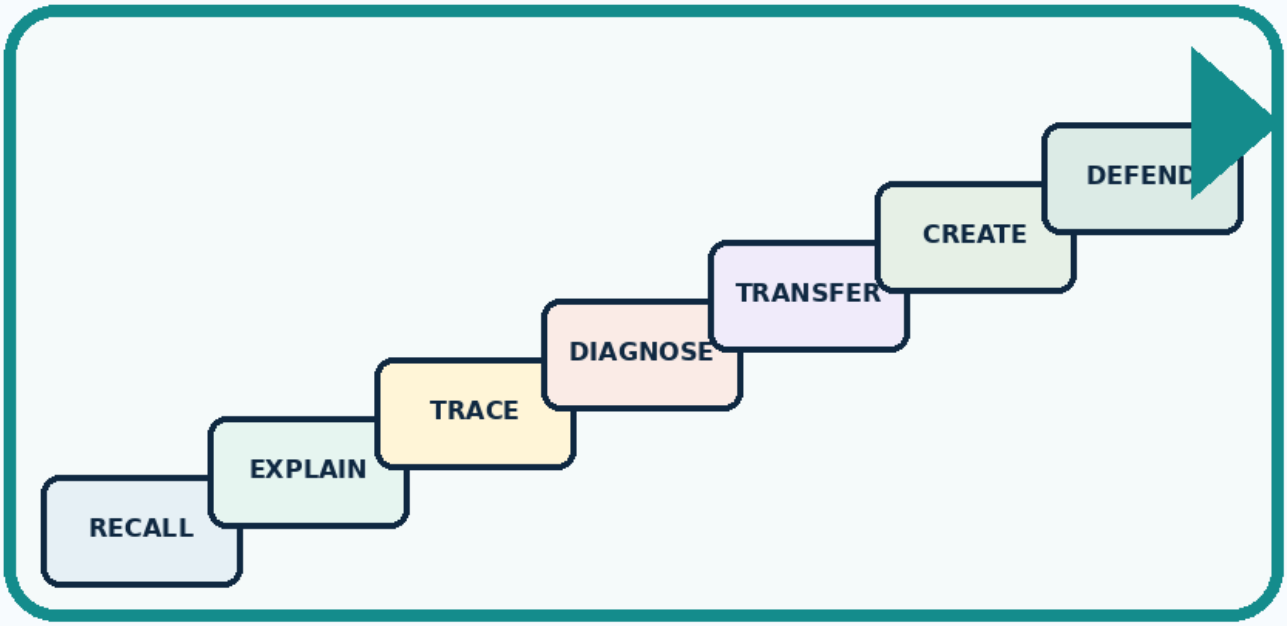
- RECALL Define **category**.
- EXPLAIN **Label**s are human choices that may **overlap** or be unclear.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Every object has exactly one correct **label**.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer the method to a new record is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer the method to a new record is the selected independent case.



EXPLANATION CHECK State the policy and its version, apply its known-failure priority and boundary, and distinguish unknown information from failure.

Independent **evidence**:

SELF-CHECK Strong **label**s include a purpose and a rule for membership. **Overlap**ping objects may need multiple **label**s or a priority rule. A revised definition should reduce **disagreement** without pretending every case is simple.

NEXT BRIDGE Carry today's **evidence** habit into: Features: Useful Clues.

▼ **Extra practice: Use the same label rule for every card**

REFERENCE · SECTION 13

Extra practice: Use the same label rule for every card

> Try the question before opening its worked explanation.

Read the supplied case and question

Six cards show two circles, two triangles and two squares. Sort by shape.

Why can red and blue circles share a **label**?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

Evidence or stage	Value or result
Circles	2
Triangles	2
Squares	2

▼ Earlier wording, alternatives and picture transcripts

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Use the same label rule for every card

Six cards show two circles, two triangles and two squares. Sort by shape.

Why can red and blue circles share a **label**?

Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Apply a category definition using explicit evidence and boundary rules	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a known failure from information that is missing or unreadable	Explain the deciding step, show a trace or test, and state one limit.
Use disagreement to improve a labelling guide without mistaking agreement for truth	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional craft club **label**s cardboard-order cards ready, review or revise.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft club **label**s cardboard-order cards ready, review or revise.
- Which **label**led step matches this detail? 2 is valid but colour is blank: review.
- 1. Read policy
- 2. Card B
- 3. Card A
- 4. Card C
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Card C 2 is valid but colour is blank: review. Operational **label**s specify what a word means for this task.

Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong. Revise is supported by a known failure.

Changed-case question

- Changed-case practice: Change B's quantity from 5 to 4; leave colour blank. Which conclusion is supported by the supplied facts?
- 1. B becomes review.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: B becomes review. The known failure is removed, but a required field is still missing.

Repair and save your work

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used **label**s and which stage used only the new measurement..

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

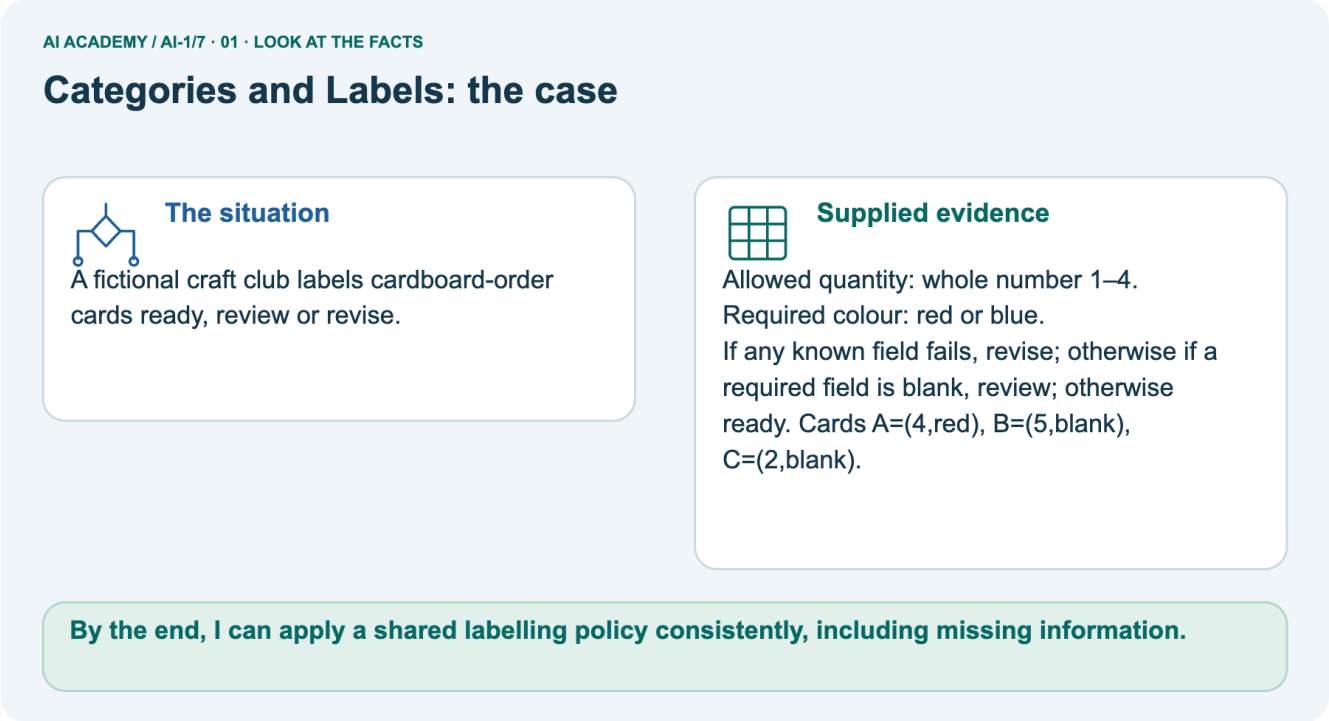
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Categories and Labels: the case



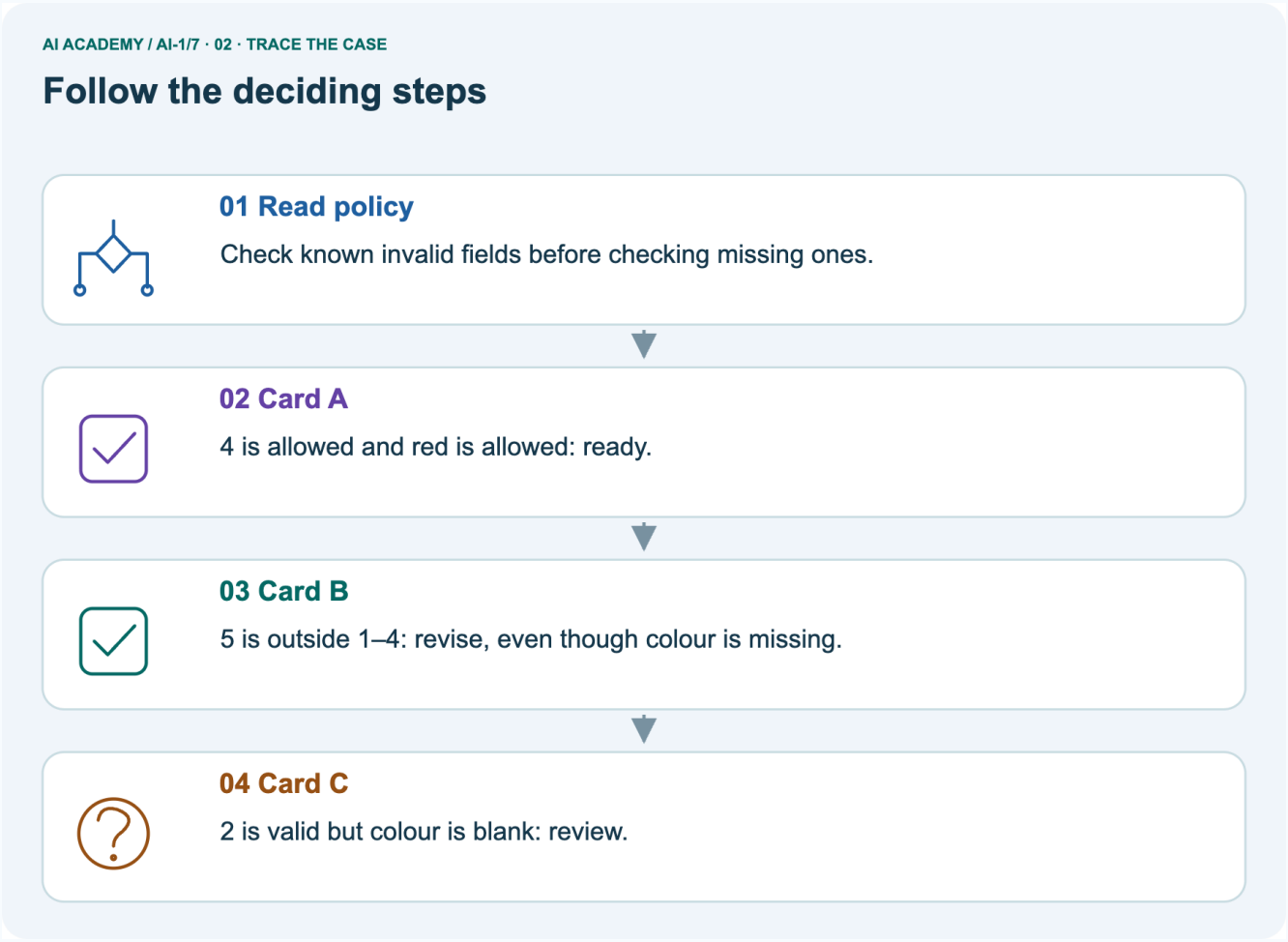
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft club **label**s cardboard-order cards ready, review or revise. Supplied facts: Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read policy: Check known invalid fields before checking missing ones. → Card A: 4 is allowed and red is allowed: ready. → Card B: 5 is outside 1–4: revise, even though colour is missing. → Card C: 2 is valid but colour is blank: review.

Point to each numbered stage and explain which recorded **input** it uses.

03. Three ordered checks for a calendar card

AI ACADEMY / AI-1/7 · 03 · INSPECT THE LESSON MODEL

Three ordered checks for a calendar card

Check in this order	Evidence	Recorded label
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Follow the labelled picture with the matching explanation. Open the vector diagram

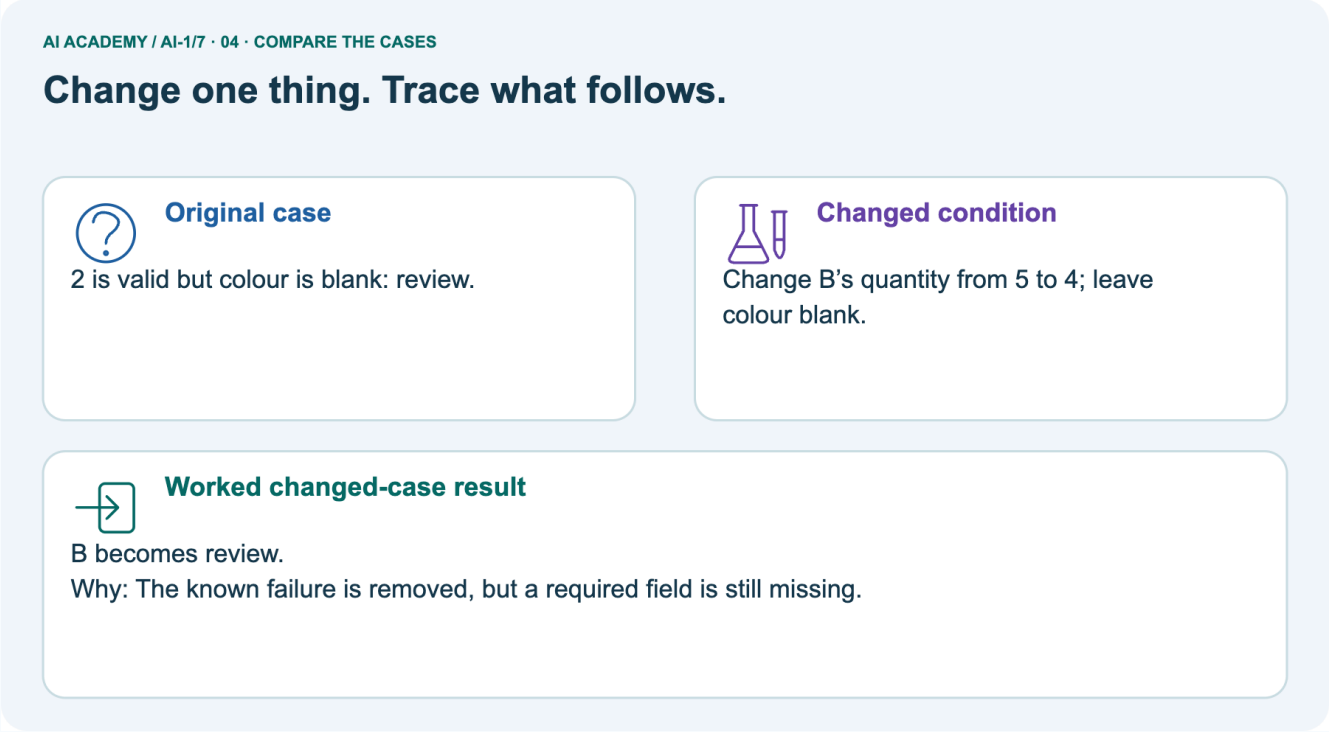
Three ordered checks for a calendar card. First check known values | Any known field fails its criterion | Revise even if another field is unknown.

Then check missing information | No known failure but a field is blank or unreadable | Review because **evidence** is incomplete.

Otherwise all checks have passed | Day is 1 through 30 and room is A or B | Ready under this record policy

Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



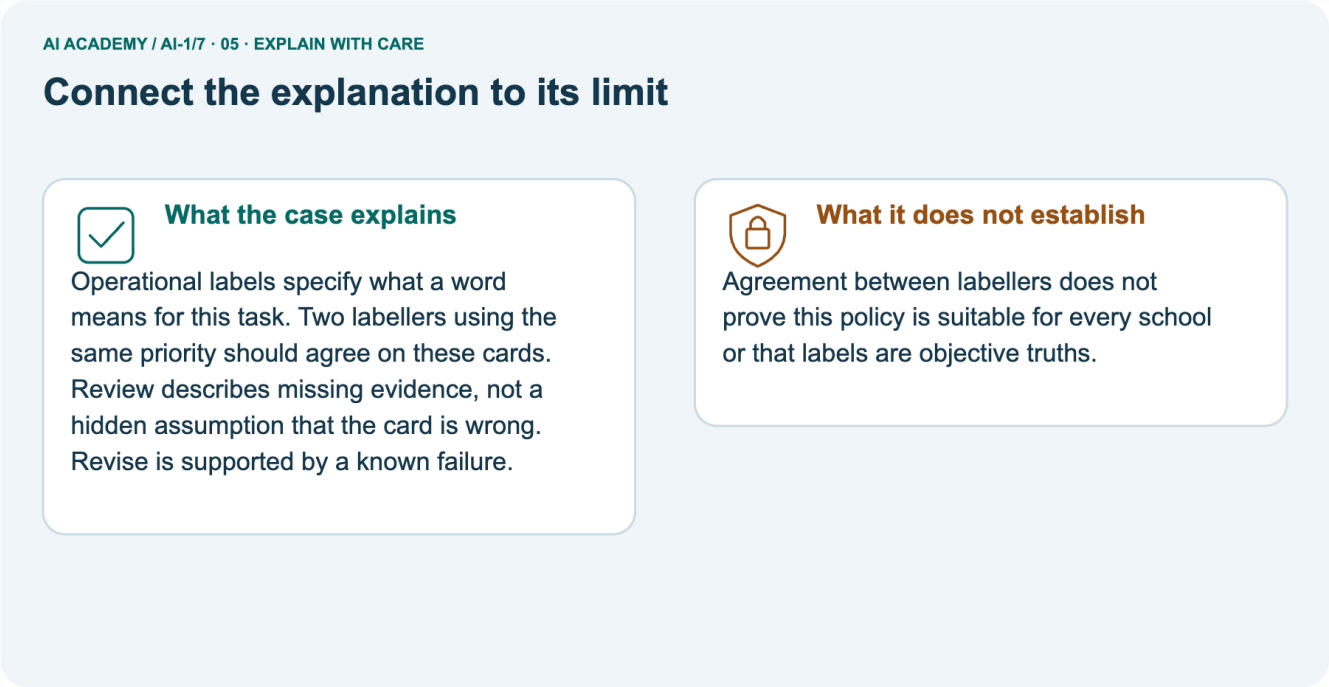
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 2 is valid but colour is blank: review. Changed condition: Change B's quantity from 5 to 4; leave colour blank. Expected result: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



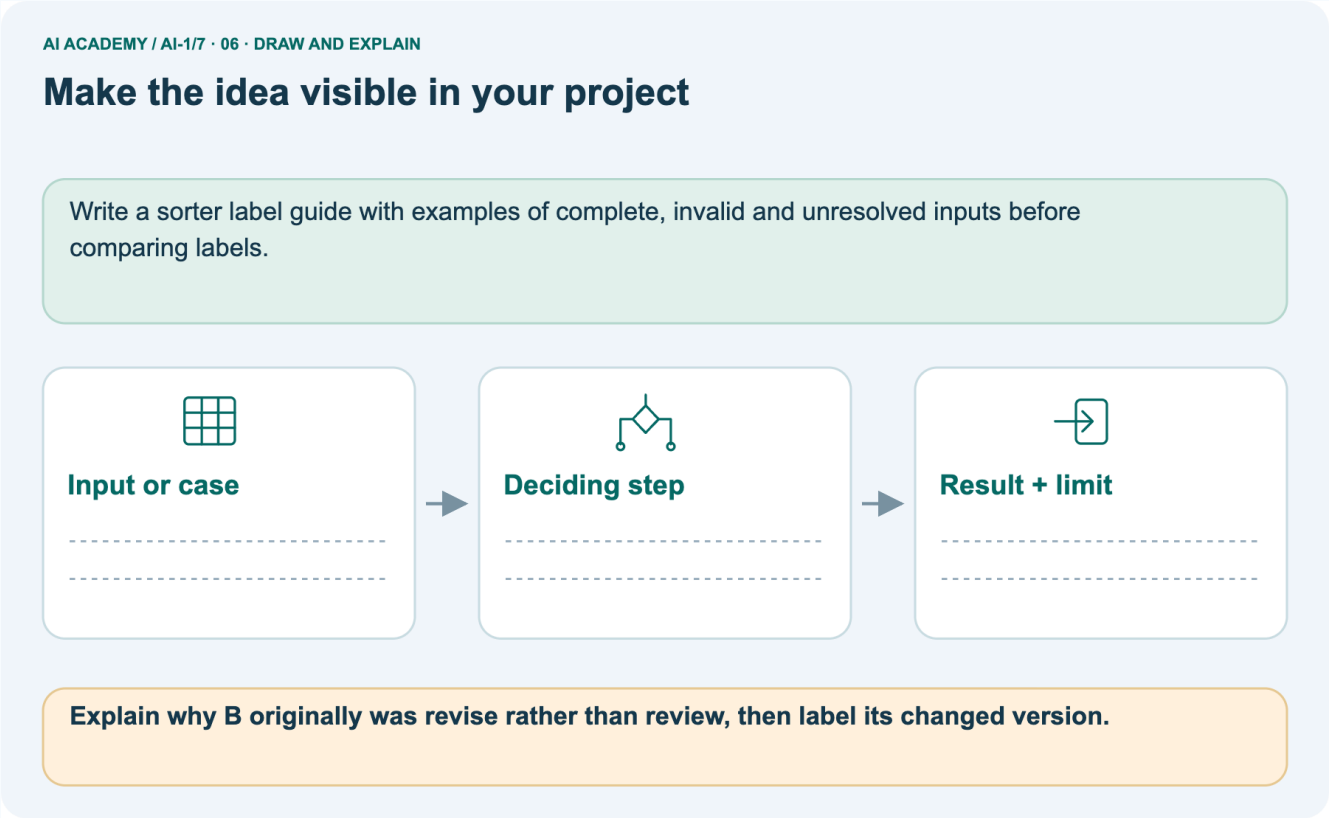
Follow the labelled picture with the matching explanation. Open the vector diagram

Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure. Limit: Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B originally was revise rather than review, then **label** its changed version.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Categories and Labels: the case

02. Follow the deciding steps

03. Three ordered checks for a calendar card

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.